Md Riad Khan

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Riad Khan | linkedin.com/in/riad-khan-a2381764 | Faculty Profile

EDUCATION

Bangladesh Maritime University (BMU) Dhaka, Bangladesh
Faculty of Engineering and Technology Apr. 2024 – Apr. 2026
M.Sc. in Naval Architecture and Offshore Engineering CGPA: 3.91/4.00

- **Relevant Courses**: Advanced Numerical Methods, Advanced Marine Hydrodynamics, Dynamics of Ships and Offshore Structure, Offshore Renewable Energy (Curriculum Link)
- **Thesis**: Design Optimization and Performance Evaluation of a Shell and Tube Heat Exchanger for Marine Cooling Applications
  - Numerically evaluated the thermal performance and erosion characteristics of an STHX by simulating realistic seawater conditions. Proposed an optimized STHX design and a cost-effective filtration strategy for marine cooling applications

Bangladesh University of Engineering and Technology (BUET) Dhaka, Bangladesh
Faculty of Mechanical Engineering (ME) May. 2012 – Feb. 2017
B.Sc. in Naval Architecture and Marine Engineering (NAME) CGPA: 3.77/4.00 (Class Position: 3/57)

- **Relevant Courses**: Fluid Mechanics, Electrical Engineering Principles, Basic Thermal Engineering, Heat Transfer, Marine Hydrodynamics, Finite Element Method, Theory of Machines, Computational Fluid Dynamics (CFD), Theory of Hydrofoils, Computer Programming (Curriculum Link)
- **Thesis**: Energy Study: Contra-rotating Propeller System
  - Developed a computer program to calculate the self and mutually induced velocities of a CRP system and predicted its efficiency using both Blade Element Theory and Circulation Theory. Compared the predicted CRP efficiency with that of conventional single-screw propeller systems

AREAS OF INTEREST

- Robotics & Autonomous Vehicles
- Artificial Intelligence & Machine Learning
- Heat Transfer
- Computational Fluid Dynamics
- Mechanics of Structure
- Engines and Combustion

RESEARCH & TEACHING EXPERIENCE

Bangladesh Maritime University (BMU) Dhaka, Bangladesh
Lecturer, Department of Naval Architecture and Offshore Engineering (NAOE) Sep. 2023 – Present

- **Offered Courses**: Mechanics of Structure, Engines and Fuels, Hydrostatics and Stability, Introduction to NAOE, Mechanical Engineering Drawing Lab, Ship Design Lab 1
- **Undergraduate Thesis Supervision**: Supervised Undergraduate theses on Robotics, Autonomous Vehicles, Image Processing, Combustion Processes, etc.
- **Reviewer**: Journal of Engineering and Technology
- **Club Coordinator**: BMU Cultural Club

SELECTED PUBLICATIONS

Journal Articles

- **Khan, M. R.**, Kundu, R., Shah, R. I., & Hasan, R. (2025). Predicting a vessel's trajectory: A simple machine learning method. *Amesia*, 6(2), 122129. https://doi.org/10.54559/amesia.1764517 (Link)
- Kundu, R., **Khan, M. R.**, Rahman, S., & Saha, G. K. (2024). Marine heat waves in a low-lying country: Occurrence, computation and impact. *Journal of Naval Architecture and Marine Engineering*, 21(2), 117124. https://doi.org/10.3329/jname.v21i2.75296 (Link)

- **Khan, M. R.**, Sharmin, N., & Hasan, S. M. R. (2025). Mobile wave crawler for energy generation: Preliminary feasibility and opportunities for Bangladesh. *Journal of Bangladesh Academy of Sciences*, 49(2), 215225. https://doi.org/10.3329/jbas.v49i2.81329 (Link)
- Talokder, M. A. R., Rahman, M. M., Haque, M. M.,**Khan, M. R.** (2022). Compatibility of the pusher-barge system in inland and coastal waterways transportation system of Bangladesh. *Journal of Maritime Research*, 19(2), 3943 (Link)

Journal Articles Under Review/ Preparation

- **Khan, M. R.**, Hasan Roni, M. R., & Karim, M. M. Numerical investigation of the impact of suspended solids in seawater on a shell and tube heat exchanger in marine cooling systems and development of a cost-effective filtration system
- **Khan, M. R.**, Hasan Roni, M. R., & Karim, M. M. Numerical Assessment of Advanced Baffle Arrangements for Improved Heat Transfer and Flow Distribution in a Shell-and-Tube Heat Exchanger
- Khan, M. M., Abir, M. J. R., **Khan, M. R.**, & Islam, S. Deep learning framework for morphological identification of Hilsa in high-turbidity riverine ecosystems
- Hasan, S. M. R., Karim, M. M., Hossain, K. A.,**Khan, M. R.**, Islam, M. S., & Raihan, A. CFD-based validation of fuel-saving ship hull design methods for inland ships. Accepted in *Scientific Reports* (PDF)

Conference Proceedings

- **Khan, M. R.**, Kundu, R., Rahaman, W., Haque, M. M., & Ullah, M. R. (2018). Efficiency study: Contra-rotating propeller system. In *AIP Conference Proceedings* (Vol. 1980, No. 1, Article 060001). International Conference on Mechanical Engineering (ICME 2017), Dhaka, Bangladesh. https://doi.org/10.1063/1.5044369 (Link)
- Kundu, R., **Khan, M. R.**, Rahman, S., & Saha, G. K. (2018). Design and structural analysis of a pontoon vessel. In *AIP Conference Proceedings* (Vol. 1980, No. 1, Article 030006). International Conference on Mechanical Engineering (ICME 2017), Dhaka, Bangladesh. https://doi.org/10.1063/1.5044285 (Link)
- Rahman, W., Sworan, M. R. I., **Khan, M. R.**, & Baree, M. S. (2018). Structural analysis of a container vessel hatch cover by finite element method. In *AIP Conference Proceedings* (Vol. 1980, No. 1, Article 030005). International Conference on Mechanical Engineering (ICME 2017), Dhaka, Bangladesh. https://doi.org/10.1063/1.5044284 (Link)
- Khaled, M. E., Kawamura, Y., Abdullah, M. S. B., Banik, A., Faruk, A. K., & **Khan, M. R.** (2018). Assessment of collision & grounding risk at Chittagong Port, Bangladesh. In *Proceedings of the 11th International Conference on Marine Technology*, Kuala Lumpur, Malaysia. (PDF)
- **Khan, M. R.**, Islam, M. A., & Abdullah, A. (2020). Time and cost effective ship design process using single parent design approach with considerable dimension difference. In *Proceedings of the 12th International Conference on Marine Technology*, Indonesia.
- Abdullah, A., Moinuddin, A., **Khan, M. R.**, & Islam, M. A. (2020). Advancement of ship recycling industry of Bangladesh in comparison to the rest of the world with obstacles, root causes and propositions. In *Proceedings of the 12th International Conference on Marine Technology*, Indonesia.

PROJECTS/ RESEARCH GRANTS

- Project Director: **Assessing Domestic Industrial Capabilities to Support Shipbuilding Industries in Bangladesh**, (Jan. 2026 - Present), Funded by: BMU
- Consultant Team Member: **Feasibility Study for the Procurement and Installation of 4 Winch Barges with Ancillary Equipment to Increase the Salvage Capacity of the Salvage Unit**, (Aug. 2025 - Jan. 2026), Funded by: Bangladesh Inland Water Transport Authority (BIWTA)
- Consultant Team Member: **Feasibility Study for Procurement of Dredge Material Carrying Dumb Barges**, (Jun. 2024 - Feb. 2025), Funded by: BIWTA
- Associate Investigator: **Sustainable Inland Shipping: Validating Fuel-Saving Ship Hull Design Methods for Inland Ships of Bangladesh**, (Jun. 2024 - Jun. 2025), Funded by: Ministry of Science and Technology, Bangladesh

TECHNICAL SKILLS

- **Programming Languages:** Python, MATLAB, Fortran, C++
- **Simulation:** ANSYS Fluent, ANSYS Mechanical, Maxsurf, SHIPFLOW
- **Design Software:** SolidWorks, Rhino 3D, AutoCAD

PROFESSIONAL EXPERIENCE

Mongla Port Authority — Mongla, Bagerhat, Bangladesh
Assistant Engineer — Jun. 2022 – Sept. 2023

- **Key Responsibilities**: Project Supervision, Troubleshooting, Design, Drafting, Estimation, Maintenance

Khulna Shipyard Ltd (KSY), Bangladesh Navy — Khulna, Bangladesh
Assistant Engineer — Jul. 2017 – Jun. 2022

- **Key Responsibilities**: Project Management and Leadership; Design, Drafting, and Planning; Calculations and Cost Estimation; Tender Preparation and Procurement; Documentation and Presentation; Quality Assurance, Control, and Supervision; Commissioning, Testing, and Trials; Technical Review, Troubleshooting, and Problem Solving
- **Major Completed Projects**: Three Inshore Patrol Vessels (IPV) for Bangladesh Coast Guard; Two Survey Vessels for Bangladesh Navy

HONORS AND AWARDS

- **University Merit Scholarship:** Awarded for outstanding academic performance in the University Examinations (2012, 2013, 2015, and 2017), BUET
- **Dean's List:** Awarded for obtaining a CGPA above 3.75 (2012, 2013, 2015, 2016), Faculty of Mechanical Engineering, BUET
- **Higher Secondary Certificate (HSC) Talent Pool Scholarship:** Awarded for outstanding performance in the HSC Examination under the Jessore Education Board, Bangladesh (2011 - 2015)
- **Appreciation Letter:** Received multiple Appreciation Letters for successful completion of projects at KSY

TRAININGS

- Training on **SHIPFLOW** by Flowtech Int. (Oct. 2025), BMU
- Smart Port Management: Empowering Economic Prosperity with **AI-Based Port** Research Bootcamp by Dr Burak Cankaya, Associate Professor, Embry-Riddle Aeronautical University, USA (7 - 11 Sep. 2025), BMU
- Training on Competency-Based Education and Training (**CBET**), (10 - 13 Mar. 2024), BMU
- Training on **Engine, Z-Peller and Remote Control System**, (18 Nov. 2022), IHI Power Systems Co. Ltd., Japan
- Course on **Japan-Asia Youth Exchange Program in Science**, (11 - 20 Dec. 2016), Osaka University, Japan

PROFESSIONAL AFFILIATIONS

- Associate Member of **Institution of Engineers, Bangladesh (IEB)**, ID: A21349
- Life Member of **Association of Naval Architects and Marine Engineers, Bangladesh (ANAMEB)**, ID: LM106220

EXTRA-CURRICULAR ACTIVITIES

- Served as a **Drummer** in local musical bands, including Cyanic Dearth, Traum, and Surreal
- Performed as an **Acoustic Guitarist** at cultural programs and community events
- Actively participated in organizing cultural, social, and other extracurricular events
- Provided private tutoring in advanced mathematics, physics, and chemistry to high school and college students.

REFERENCES

Dr. Md. Sadiqul Baree
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